

LAB EXPERIMENTS

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SET 6: Product Sales Analysis

Question 1: Grouped Bar Chart

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  Product = c("Product A", "Product B", "Product C"),
```

```
  January = c(2000, 1500, 1200),
```

```
  February = c(2200, 1800, 1400),
```

```
  March = c(2400, 1600, 1100)
```

```
)
```

```
sales_long <- pivot_longer(
```

```
  sales,
```

```
  cols = c(January, February, March),
```

```
  names_to = "Month",
```

```
  values_to = "Sales"
```

```
)
```

```
ggplot(sales_long,
```

```
  aes(x = Product,
```

```
    y = Sales,
```

```
    fill = Month)) +
```

```
geom_bar(stat = "identity",
```

```
  position = "dodge") +
```

```
labs(
```

```

title = "Quarter 1 Product Sales",
x = "Product",
y = "Sales ($)"
) +
theme_minimal()

```

Output



Question 2: Stacked Area Chart

```
library(ggplot2)
```

```
library(tidyr)
```

```
# Dataset
```

```

sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

```

```
# Convert to long format
```

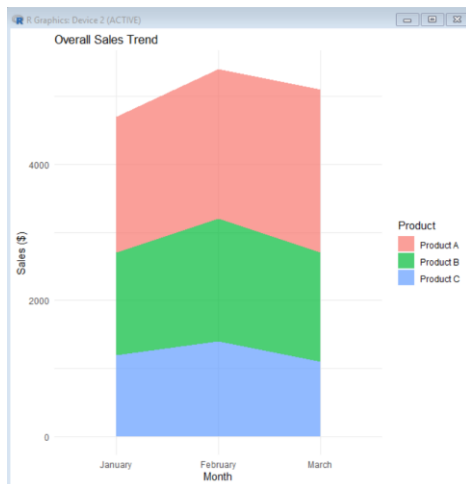
```

sales_long <- pivot_longer(
  sales,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)

# Arrange month order
sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January", "February", "March")
)

# Stacked Area Chart
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product,
    group = Product)) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales ($)"
  ) +
  theme_minimal()

```



Question 3: Sales Table

```
sales <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January_Sales = c(2000,1500,1200),
  February_Sales = c(2200,1800,1400),
  March_Sales = c(2400,1600,1100)
)
```

Display Table

```
print("Monthly Product Sales")
```

```
print(sales)
```

Output

Monthly Product Sales

```
> print(sales)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A          2000           2200          2400
2          2   Product B          1500           1800          1600
3          3   Product C          1200           1400          1100
> |
```

Question 4: Dashboard in R (Instead of Tableau)

```
library(shiny)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
sales_long <- pivot_longer(  
  sales,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)
```

```
ui <- fluidPage(  
  titlePanel("Product Sales Dashboard"),  
  sidebarLayout(  
    sidebarPanel(  
      selectInput(  
        "product",  
        "Select Product",  
        choices = c("All", sales$Product)  
      )  
    ),  
    mainPanel(  

```

```
plotOutput("bar"),
```

```
plotOutput("area"),
```

```
tableOutput("table")
```

```
)
```

```
)
```

```
)
```

```
server <- function(input, output){
```

```
  data <- reactive({
```

```
    if(input$product=="All")
```

```
      sales_long
```

```
    else
```

```
      subset(sales_long, Product==input$product)
```

```
  })
```

```
  output$bar <- renderPlot({
```

```
    ggplot(data(),
```

```
      aes(Product,Sales,fill=Month))+
```

```
    geom_bar(stat="identity",position="dodge")+
```

```
labs(title="Grouped Bar Chart")
```

```
})
```

```
output$area <- renderPlot({
```

```
  ggplot(data(),
```

```
    aes(Month,Sales,fill=Product,
```

```
    group=Product))+
```

```
  geom_area(alpha=0.7)+
```

```
  labs(title="Stacked Area Chart")
```

```
})
```

```
output$table <- renderTable({
```

```
  if(input$product=="All")
```

```
    sales
```

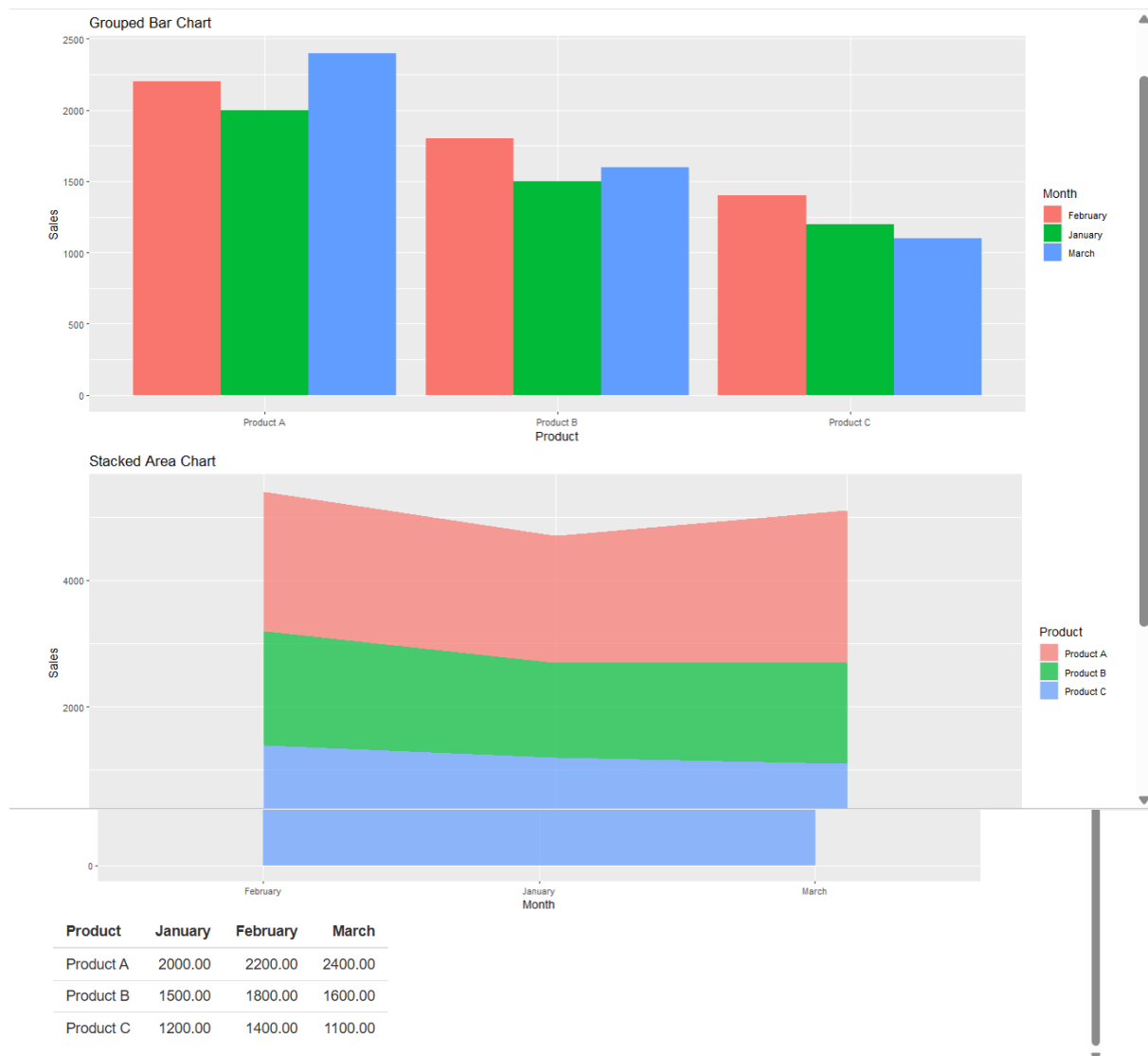
```
  else
```

```
    subset(sales, Product==input$product)
```

```
})
```

```
}
```

```
shinyApp(ui,server)
```



SET 7: Customer Demographics Analysis

Question 1: Bar Chart – Customer Ages

```
library(ggplot2)
```

```
# Dataset
```

```
customer <- data.frame(
```

```
  CustomerID = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

```
# Bar Chart
```

```
ggplot(customer, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Customer Age Distribution",  
    x = "Customer ID",  
    y = "Age (Years)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

Output



Question 2: Pie Chart – Gender Distribution

```
library(ggplot2)
```

```
# Dataset
```

```
customer <- data.frame(  
  CustomerID = c(1,2,3),
```

```

Age = c(28,35,42),
Gender = c("Female","Male","Female"),
Income = c(50000,60000,75000)
)

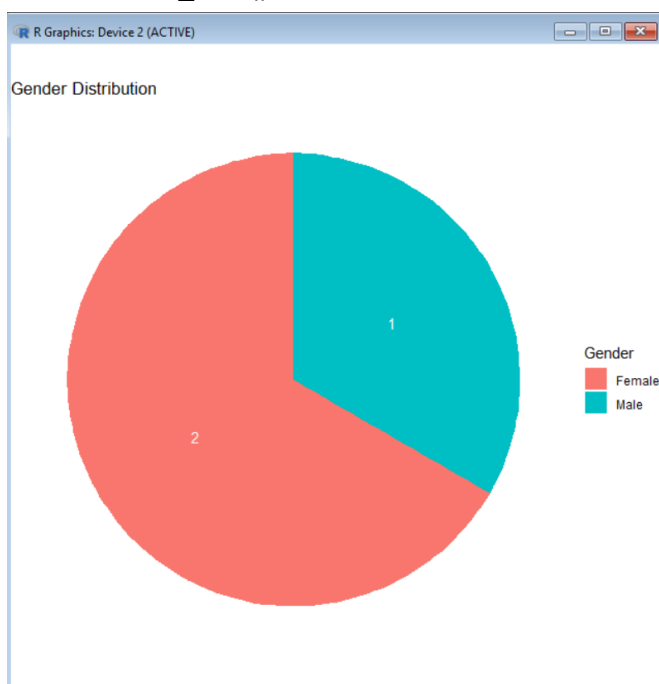
# Count genders
gender_data <- as.data.frame(table(customer$Gender))
colnames(gender_data) <- c("Gender", "Count")

```

```

# Pie Chart
ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  geom_text(aes(label = Count),
            position = position_stack(vjust = 0.5),
            color = "white") +
  labs(title = "Gender Distribution") +
  theme_void()

```



Output

Question 3: Customer Demographics Table

```
customer <- data.frame(  
  Customer_ID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
  
cat("Customer Demographics Information\n\n")  
  
print(customer)
```

Output

```
> print(customer)  
  Customer_ID Age Gender Income  
1           1  28 Female  50000  
2           2  35   Male  60000  
3           3  42 Female  75000  
.
```

Question 4: Dashboard in R (Shiny Instead of Tableau)

```
library(shiny)  
library(ggplot2)
```

Dataset

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Customer Demographics Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput(
```

```
        "gender",
```

```
        "Select Gender",
```

```
        choices = c("All", unique(customer$Gender))
```

```
      )
```

```
    ),
```

```
    mainPanel(
```

```
      plotOutput("barChart"),
```

```
      plotOutput("pieChart"),
```

```
      tableOutput("tableData")
```

```
    )
```

)

)

```
server <- function(input, output){
```

```
  data <- reactive({
```

```
    if(input$gender == "All")
```

```
      customer
```

```
    else
```

```
      subset(customer, Gender == input$gender)
```

```
  })
```

```
  output$barChart <- renderPlot({
```

```
    ggplot(data(),
```

```
      aes(x = factor(CustomerID),
```

```
        y = Age,
```

```
        fill = factor(CustomerID))) +
```

```
    geom_bar(stat = "identity") +
```

```
    labs(
```

```
      title = "Customer Age Distribution",
```

```
      x = "Customer ID",
```

```
      y = "Age"
```

```
    ) +
```

```
    theme_minimal() +
```

```

    theme(legend.position = "none")

  })

output$pieChart <- renderPlot({

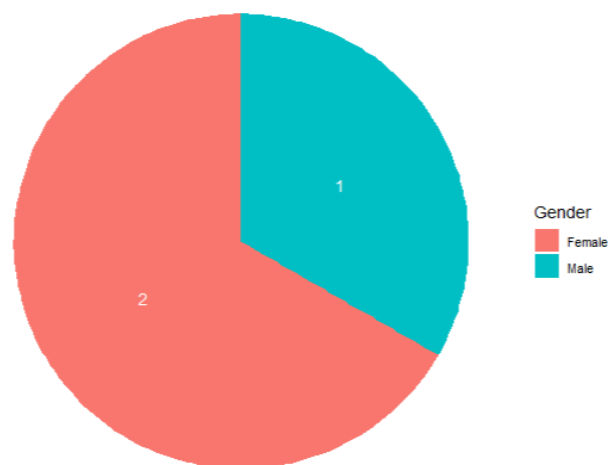
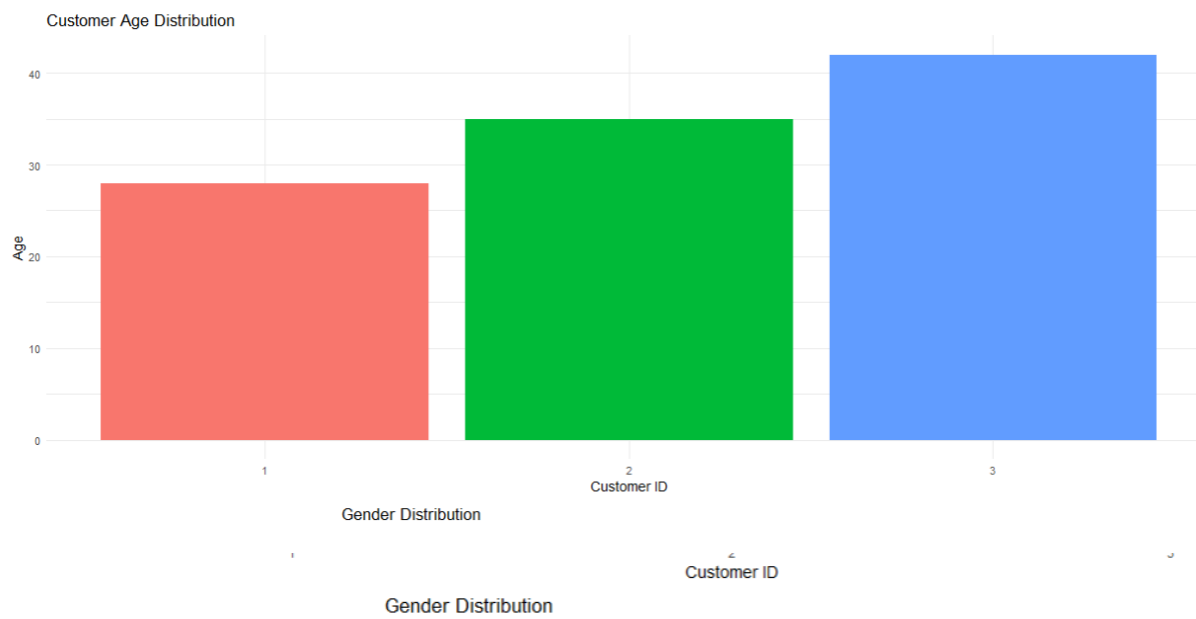
  pie_data <- as.data.frame(table(data()$Gender))
  colnames(pie_data) <- c("Gender", "Count")

  ggplot(pie_data,
    aes(x = "", y = Count, fill = Gender)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar("y") +
    geom_text(aes(label = Count),
      position = position_stack(vjust = 0.5),
      color = "white") +
    labs(title = "Gender Distribution") +
    theme_void()
  })

output$tableData <- renderTable({
  data()
})
}

shinyApp(ui, server)

```



CustomerID	Age	Gender	Income
1.00	28.00	Female	50000.00
2.00	35.00	Male	60000.00
3.00	42.00	Female	75000.00

SET 8: Employee Performance Analysis

Question 1: Line Chart – Employee Performance Trend

```
library(ggplot2)
```

```
# Dataset
```

```
employee <- data.frame(
```

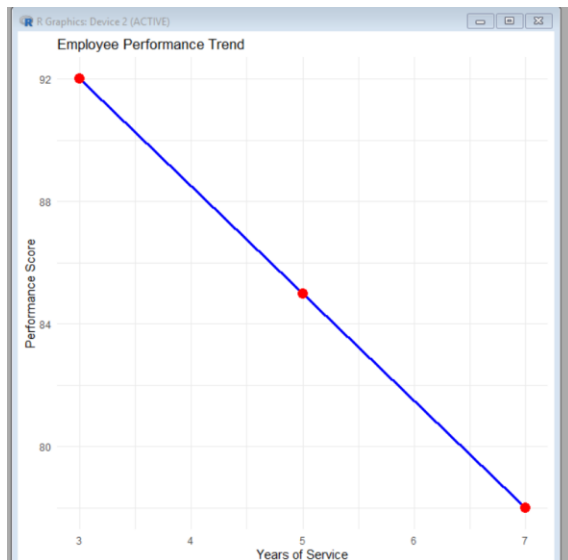
```
EmployeeID = c(1,2,3),  
Department = c("Sales","HR","Marketing"),  
YearsService = c(5,3,7),  
PerformanceScore = c(85,92,78)  
)
```

Line Chart

```
ggplot(employee,  
  aes(x = YearsService,  
      y = PerformanceScore,  
      group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(size = 4, color = "red") +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score"  
  ) +  
  theme_minimal()
```

Output

A line chart showing the employee performance trend based on years of service.



Question 2: Bar Chart – Employees by Department

```
library(ggplot2)
```

```
# Dataset
```

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

```
# Bar Chart
```

```
ggplot(employee,  
  aes(x = Department,  
    fill = Department)) +  
geom_bar() +  
labs(  
  title = "Employees Across Departments",  
  x = "Department",
```

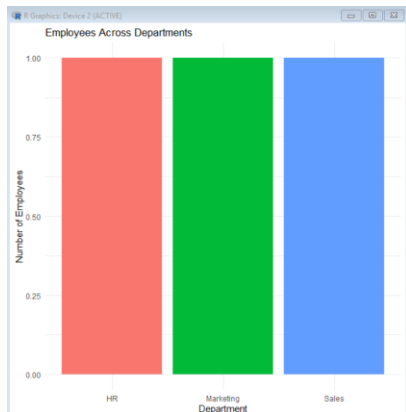
```

y = "Number of Employees"
) +
theme_minimal() +
theme(legend.position = "none")

```

Output

A bar chart showing the number of employees in each department.



Question 3: Employee Performance Table

Dataset

```

employee <- data.frame(
  Employee_ID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  Years_of_Service = c(5,3,7),
  Performance_Score = c(85,92,78)
)

```

```
cat("Employee Performance Data\n\n")
```

```
print(employee)
```

Output

Employee Performance Data

```
>
> print(employee)
Employee_ID Department Years_of_Service Performance_Score
1          1      Sales              5             85
2          2        HR              3             92
3          3  Marketing              7             78
> |
```

Question 4: Dashboard in R (Shiny Instead of Tableau)

```
library(shiny)
```

```
library(ggplot2)
```

```
# Dataset
```

```
employee <- data.frame(
```

```
  EmployeeID = c(1,2,3),
```

```
  Department = c("Sales","HR","Marketing"),
```

```
  YearsService = c(5,3,7),
```

```
  PerformanceScore = c(85,92,78)
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Employee Performance Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput(
```

```
        "dept",
```

```
        "Select Department",
```

```
        choices = c("All", unique(employee$Department))
```

```
)

),

mainPanel(

  plotOutput("lineChart"),

  plotOutput("barChart"),

  tableOutput("employeeTable")

)

)

)

server <- function(input, output){

  data <- reactive({

    if(input$dept == "All")
      employee
    else
      subset(employee, Department == input$dept)

  })
```

```
output$lineChart <- renderPlot({

  ggplot(data(),
    aes(x = YearsService,
      y = PerformanceScore,
      group = 1)) +
    geom_line(color = "blue", linewidth = 1.2) +
    geom_point(size = 4, color = "red") +
    labs(
      title = "Employee Performance Trend",
      x = "Years of Service",
      y = "Performance Score"
    ) +
    theme_minimal()

})
```

```
output$barChart <- renderPlot({

  ggplot(data(),
    aes(x = Department,
      fill = Department)) +
    geom_bar() +
    labs(
      title = "Employees Across Departments",
      x = "Department",
      y = "Number of Employees"
    )
  )
```

```

) +
theme_minimal() +
theme(legend.position = "none")
})
output$employeeTable <- renderTable({
  data()
})
}
shinyApp(ui, server)

```



SET 9 – Question 1

Histogram and Boxplot

```
library(ggplot2)
```

```
# Dataset
```

```
data <- data.frame(  
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Age = c(20, 22, 19, 21, 23, 20),  
  Course = c("R", "R", "SQL", "R", "R", "SQL"),  
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),  
  Videos_Watched = c(12, 15, 8, 18, 9, 14),  
  Quiz_Score = c(78, 85, 65, 92, 70, 88),  
  Login_Date = c("2025-01-05", "2025-01-05",  
    "2025-02-08", "2025-02-08",  
    "2025-03-12", "2025-03-12")  
)
```

```
# Histogram
```

```
ggplot(data, aes(x=Quiz_Score))+  
  geom_histogram(binwidth=5,  
    fill="skyblue",  
    color="black")+  
  labs(title="Distribution of Quiz Scores",  
    x="Quiz Score",  
    y="Frequency")+  
  theme_minimal()
```

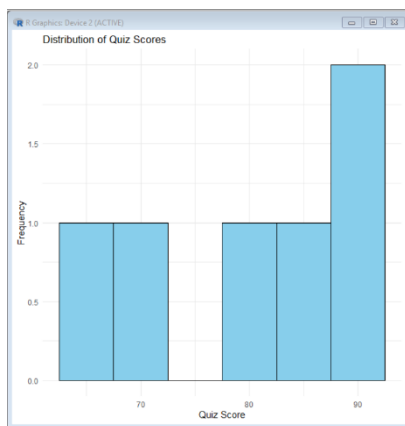
```
# Boxplot
```

```
ggplot(data, aes(Course, Quiz_Score, fill=Course))+
```

```
geom_boxplot()+
labs(title="Quiz Score by Course",
      x="Course",
      y="Quiz Score")+
theme_minimal()
```

Interpretation

- Most students scored between **70–90**.
- R course students performed slightly better than SQL students.



SET 9 – Question 2

Scatter Plot

```
library(ggplot2)

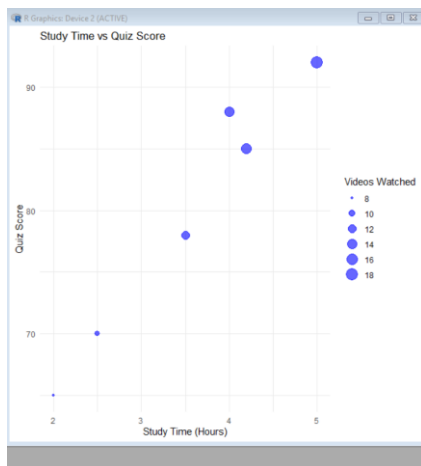
data <- data.frame(
  Student_ID=c("L01","L02","L03","L04","L05","L06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Age=c(20,22,19,21,23,20),
  Course=c("R","R","SQL","R","R","SQL"),
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched=c(12,15,8,18,9,14),
  Quiz_Score=c(78,85,65,92,70,88)
)

ggplot(data,
```

```

aes(Study_Time,
     Quiz_Score,
     size=Videos_Watched))+
geom_point(alpha=0.6,
           color="blue")+
labs(title="Study Time vs Quiz Score",
     x="Study Time (Hours)",
     y="Quiz Score",
     size="Videos Watched")+
theme_minimal()

```



SET 9 – Question 3

Monthly Average Quiz Score

```
library(ggplot2)
```

```
# Dataset
```

```

data <- data.frame(
  Student_ID=c("L01","L02","L03","L04","L05","L06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Age=c(20,22,19,21,23,20),
  Course=c("R","R","SQL","R","R","SQL"),
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),

```

```

Videos_Watched=c(12,15,8,18,9,14),
Quiz_Score=c(78,85,65,92,70,88),
Login_Date=c("2025-01-05","2025-01-05",
              "2025-02-08","2025-02-08",
              "2025-03-12","2025-03-12")
)

# Convert to Date
data$Login_Date <- as.Date(data$Login_Date)

# Month
data$Month <- format(data$Login_Date,"%Y-%m")

# Average Score
avg <- aggregate(Quiz_Score~Month,data,mean)

# Moving Average
avg$MovingAverage <- c(
  NA,
  mean(avg$Quiz_Score[1:2]),
  mean(avg$Quiz_Score[2:3])
)

# Plot
ggplot(avg,aes(Month,Quiz_Score,group=1))+
  geom_line(color="blue",linewidth=1.2)+
  geom_point(size=3)+
  geom_line(aes(y=MovingAverage),

```

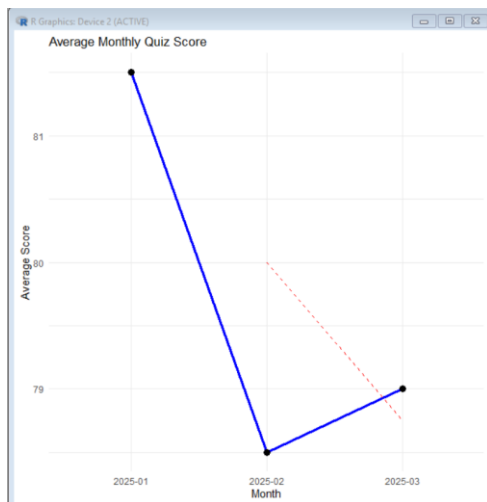
```

    color="red",
    linetype="dashed")+
labs(title="Average Monthly Quiz Score",
     x="Month",
     y="Average Score")+
theme_minimal()

```

Trend

- The moving average smooths the month-to-month fluctuations.
- Overall performance remains relatively consistent across the three months.



SET 9 – Question 4

R Shiny Dashboard (Instead of Tableau)

```
library(shiny)
```

```
library(ggplot2)
```

```
# Dataset
```

```
data <- data.frame(
```

```
  Student_ID=c("L01","L02","L03","L04","L05","L06"),
```

```
  Gender=c("Male","Female","Male","Female","Male","Female"),
```

```
  Age=c(20,22,19,21,23,20),
```

```
  Course=c("R","R","SQL","R","R","SQL"),
```

```

Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
Videos_Watched=c(12,15,8,18,9,14),
Quiz_Score=c(78,85,65,92,70,88)
)
ui <- fluidPage(
  titlePanel("Online Learning Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("course",
        "Course",
        choices=c("All",unique(data$Course)))
    ),
    mainPanel(
      plotOutput("bar"),
      plotOutput("scatter"),
      tableOutput("table")
    )
  )
)
server <- function(input,output){

  filtered <- reactive({

    if(input$course=="All")
      data
    else
      subset(data,Course==input$course)
  })
}

```

```
}}
```

```
output$bar <- renderPlot({
```

```
  avg <- aggregate(Quiz_Score~Course,  
                   filtered(),  
                   mean)
```

```
  ggplot(avg,  
    aes(Course,Quiz_Score,fill=Course))+  
  geom_bar(stat="identity")+  
  labs(title="Average Quiz Score by Course")
```

```
}}
```

```
output$scatter <- renderPlot({
```

```
  ggplot(filtered(),  
    aes(Study_Time,  
        Quiz_Score,  
        size=Videos_Watched))+  
  geom_point(alpha=0.6,color="blue")+  
  labs(title="Study Time vs Quiz Score")
```

```
}}
```

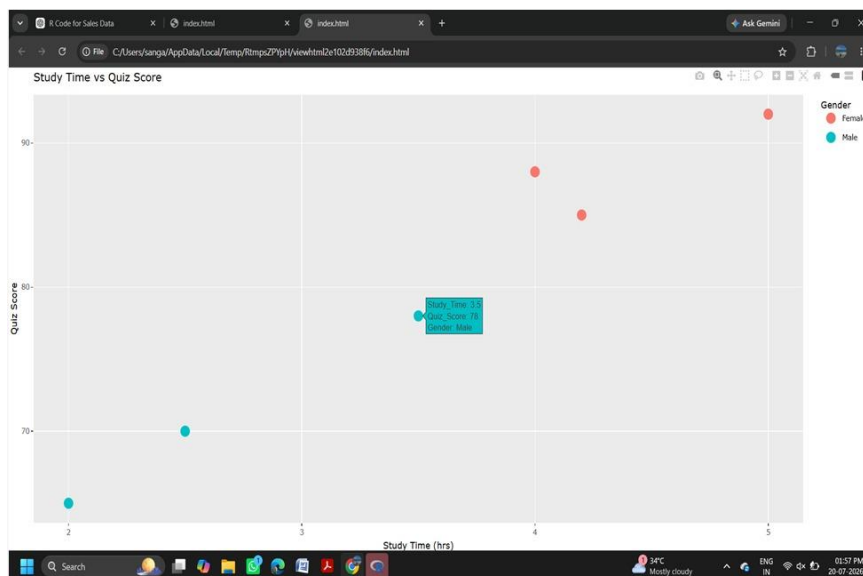
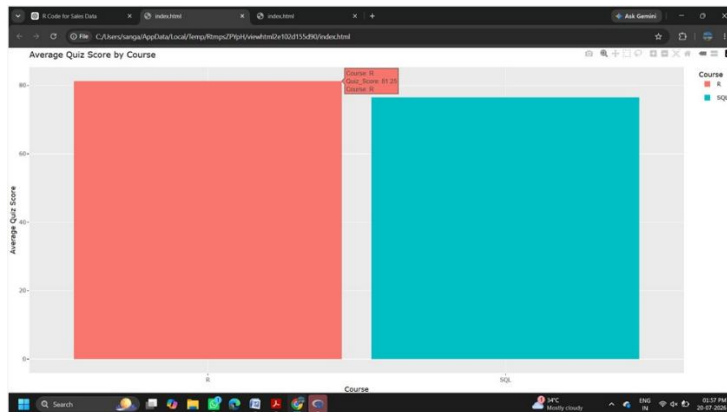
```
output$table <- renderTable({
```

```
  filtered()
```

```
}}
```

```
}
```

```
shinyApp(ui,output,server)
```



SET 10: Survey Responses Analysis

Question 1: Grouped Bar Chart

```
library(ggplot2)
```

```
# Dataset
```

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")
```

```

)

q1 <- as.data.frame(table(survey$Question1))

colnames(q1) <- c("Answer","Count")

# Grouped Bar Chart

ggplot(q1,

  aes(x=Answer,

    y=Count,

    fill=Answer))+

geom_bar(stat="identity")+

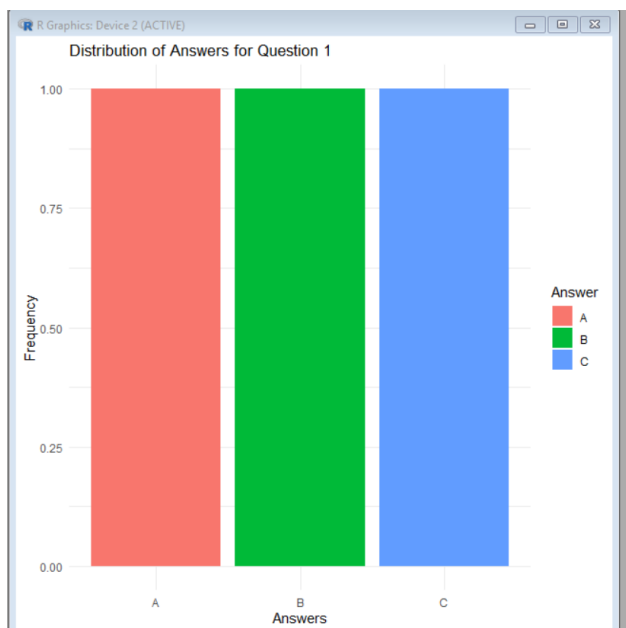
labs(title="Distribution of Answers for Question 1",

  x="Answers",

  y="Frequency")+

theme_minimal()

```



Question 2: Stacked Bar Chart

```

library(ggplot2)

library(tidyr)

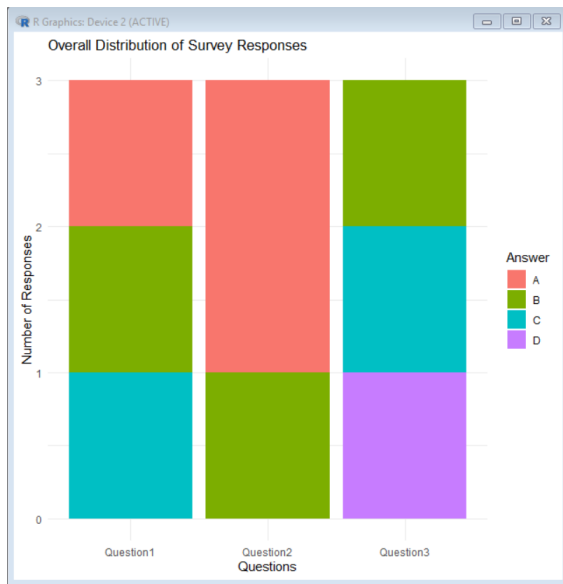
survey <- data.frame(

  SurveyID = c(1,2,3),

```

```
Question1 = c("A","B","C"),  
Question2 = c("B","A","A"),  
Question3 = c("C","D","B")  
)  
survey_long <- pivot_longer(  
  survey,  
  cols = c(Question1,Question2,Question3),  
  names_to = "Question",  
  values_to = "Answer"  
)
```

```
# Stacked Bar Chart  
ggplot(survey_long,  
  aes(x=Question,  
    fill=Answer))+  
  geom_bar()+  
  labs(title="Overall Distribution of Survey Responses",  
    x="Questions",  
    y="Number of Responses")+  
  theme_minimal()
```



Question 3: Survey Response Table

Dataset

```
survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question_1 = c("A","B","C"),
  Question_2 = c("B","A","A"),
  Question_3 = c("C","D","B")
)
```

```
cat("Survey Response Data\n\n")
```

```
print(survey)
```

Output

```
> print(survey)
  Survey_ID Question_1 Question_2 Question_3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
> |
```

Question 4: R Shiny Dashboard (Instead of Tableau)

```
library(shiny)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
# Dataset
```

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
# Convert to long format
```

```
survey_long <- pivot_longer(  
  survey,  
  cols = c(Question1,Question2,Question3),  
  names_to = "Question",  
  values_to = "Answer"  
)
```

```
ui <- fluidPage(  
  

```

```
  titlePanel("Survey Responses Dashboard"),  
  

```

```
  sidebarLayout(  
  

```

```
    sidebarPanel(  
  

```

```
      selectInput(  
  

```

```

      "question",
      "Select Question",
      choices = c("All",
                  unique(survey_long$Question))
    )

  ),

  mainPanel(

    plotOutput("groupedBar"),

    plotOutput("stackedBar"),

    tableOutput("surveyTable")
  )
)

server <- function(input, output){

  filtered <- reactive({
    if(input$question=="All")
      survey_long
    else
      subset(survey_long,
             Question==input$question)
  })

  output$groupedBar <- renderPlot({
    q1 <- as.data.frame(table(filtered())$Answer))
  })
}

```

```
colnames(q1) <- c("Answer", "Count")
```

```
ggplot(q1,  
  aes(Answer,  
    Count,  
    fill=Answer))+  
geom_bar(stat="identity")+  
labs(title="Answer Distribution")+  
theme_minimal()  
  
})
```

```
output$stackedBar <- renderPlot({  
  ggplot(filtered(),  
    aes(Question,  
      fill=Answer))+  
  geom_bar()+  
  labs(title="Survey Response Distribution")+  
  theme_minimal()  
})
```

```
output$surveyTable <- renderTable({  
  survey  
})
```

```
}
```

```
shinyApp(ui = ui, server = server)
```

